

Creative Lighting:

Programming Dynamic LED Lights with Arduino

Kingston School of Art

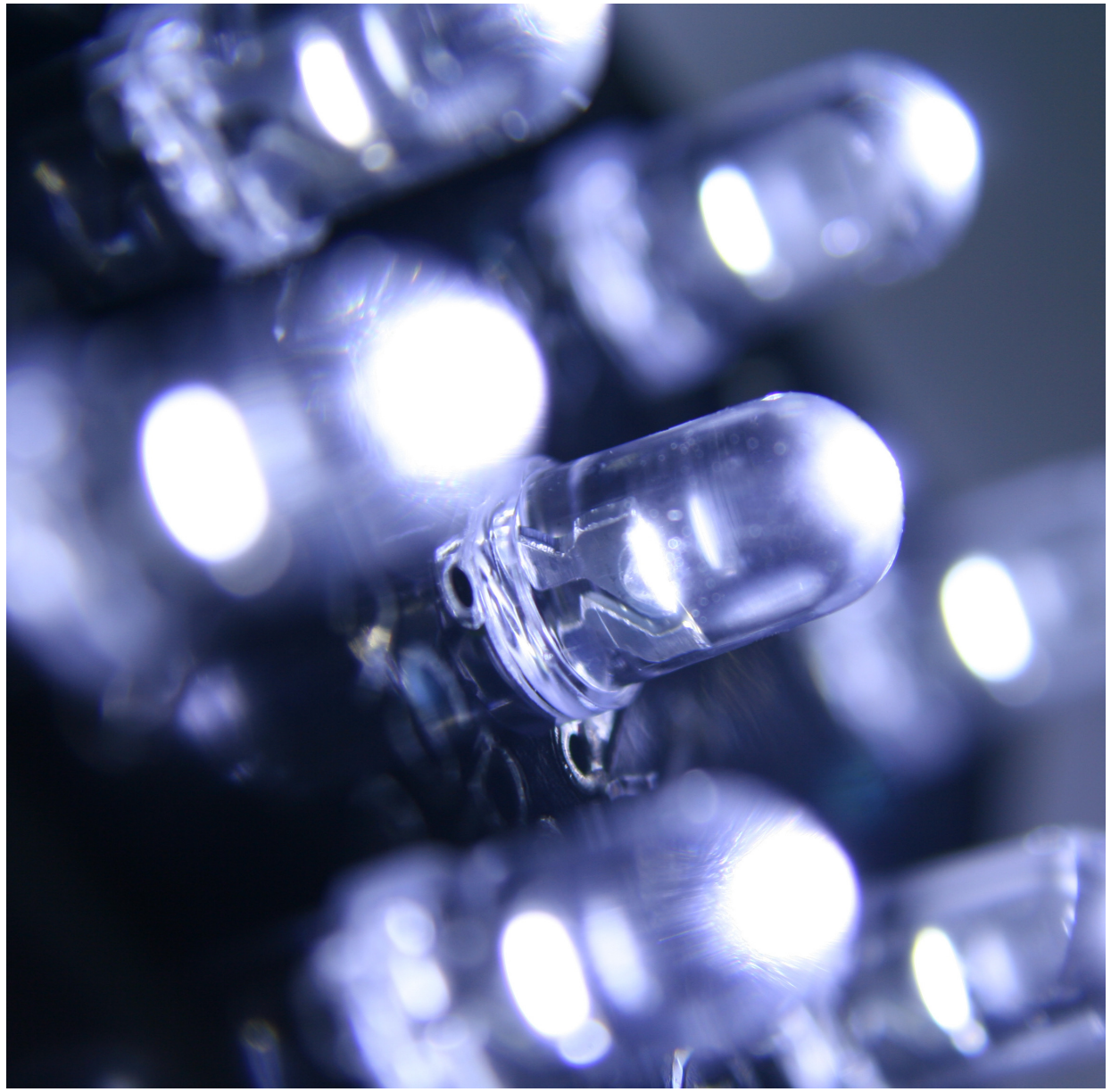
hackSpace
KSA Short Courses

LED TYPES

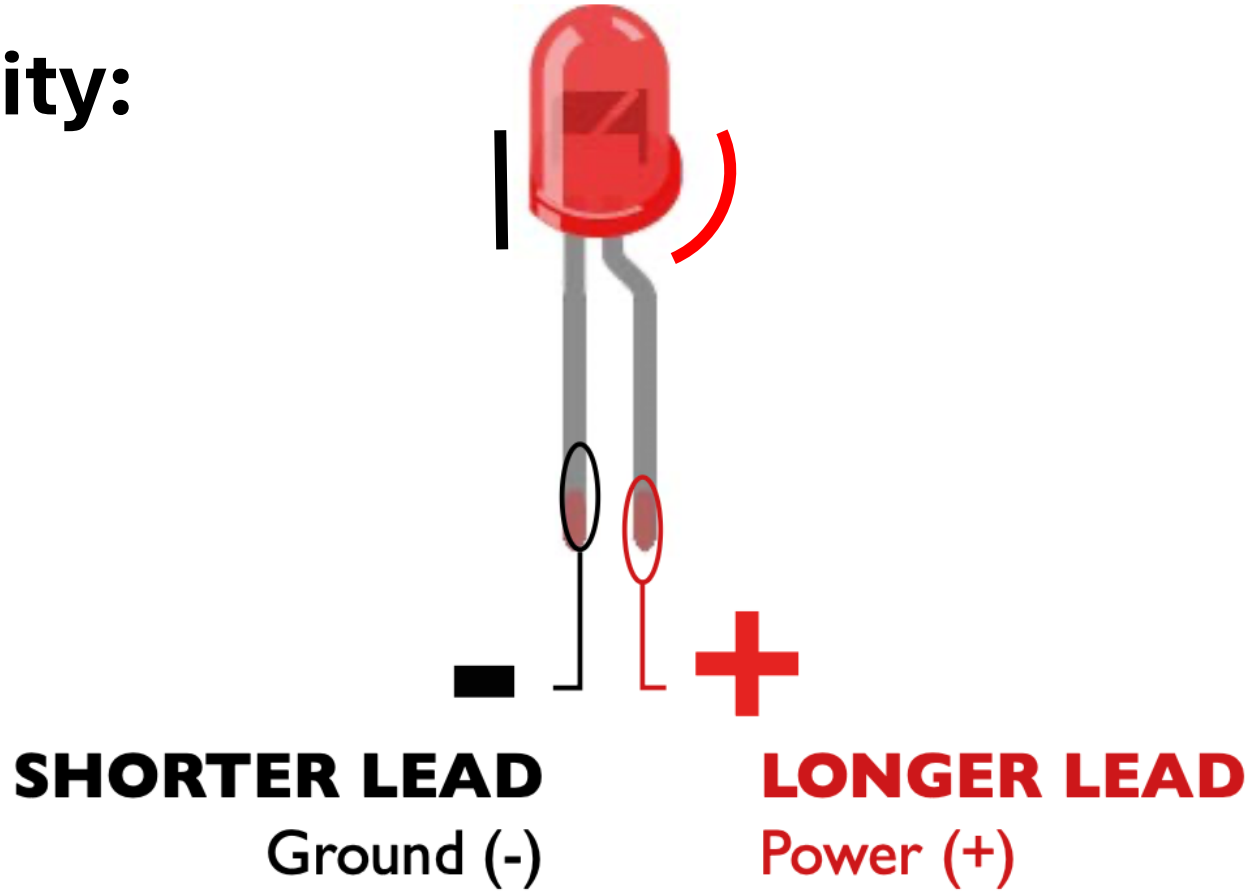
WHAT IS AN LED?

Light-Emitting-Diode (LED)

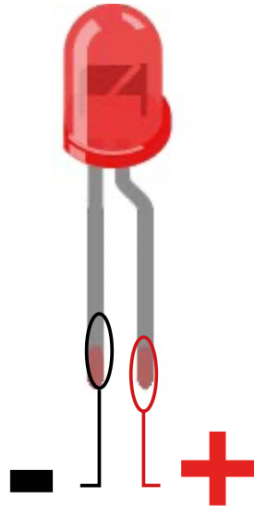
LEDs are electronic components (diodes) that emit light when electricity flows through them.



LEDs have polarity:

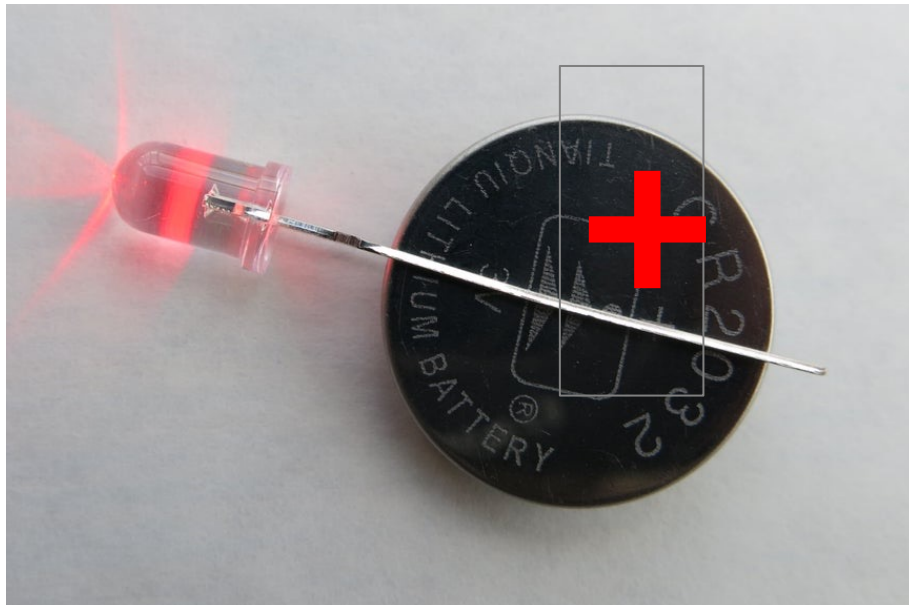


ACTIVITY



SHORTER LEAD
Ground (-)

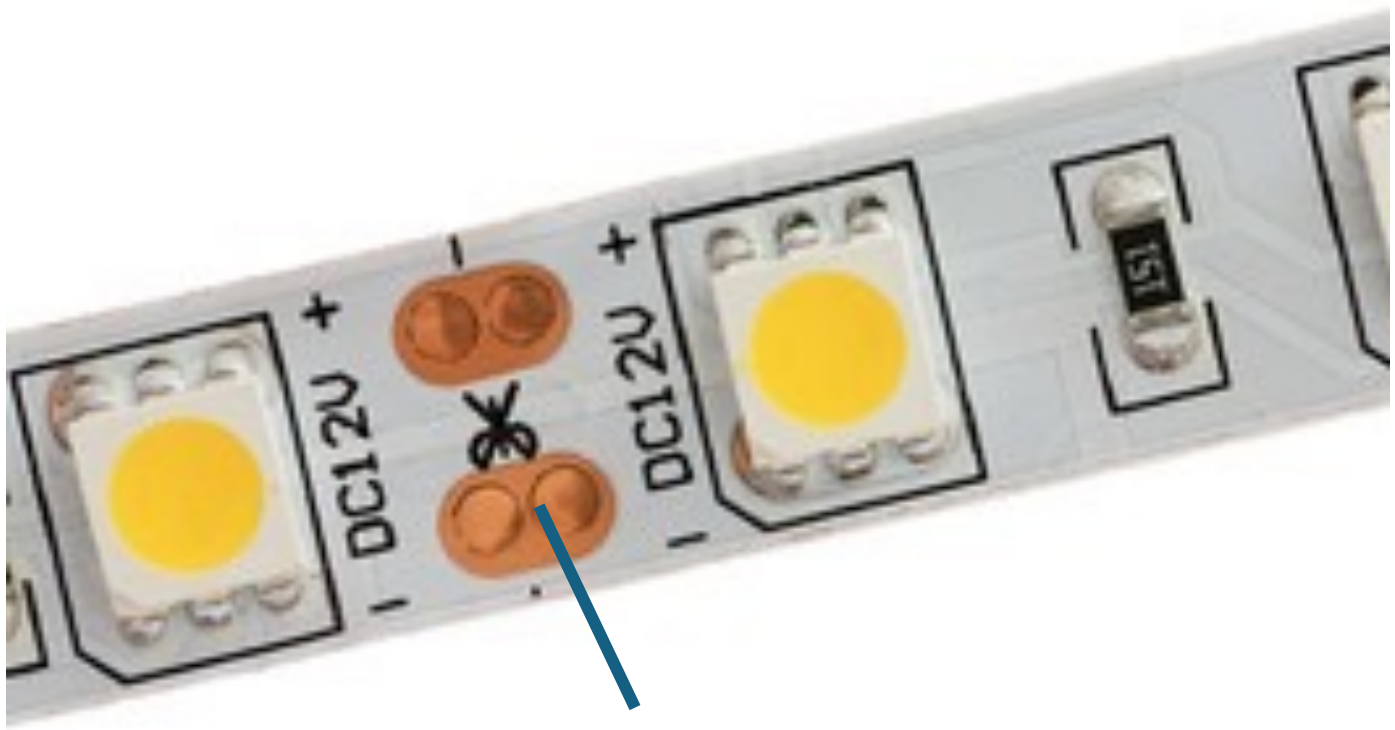
LONGER LEAD
Power (+)



LED strips



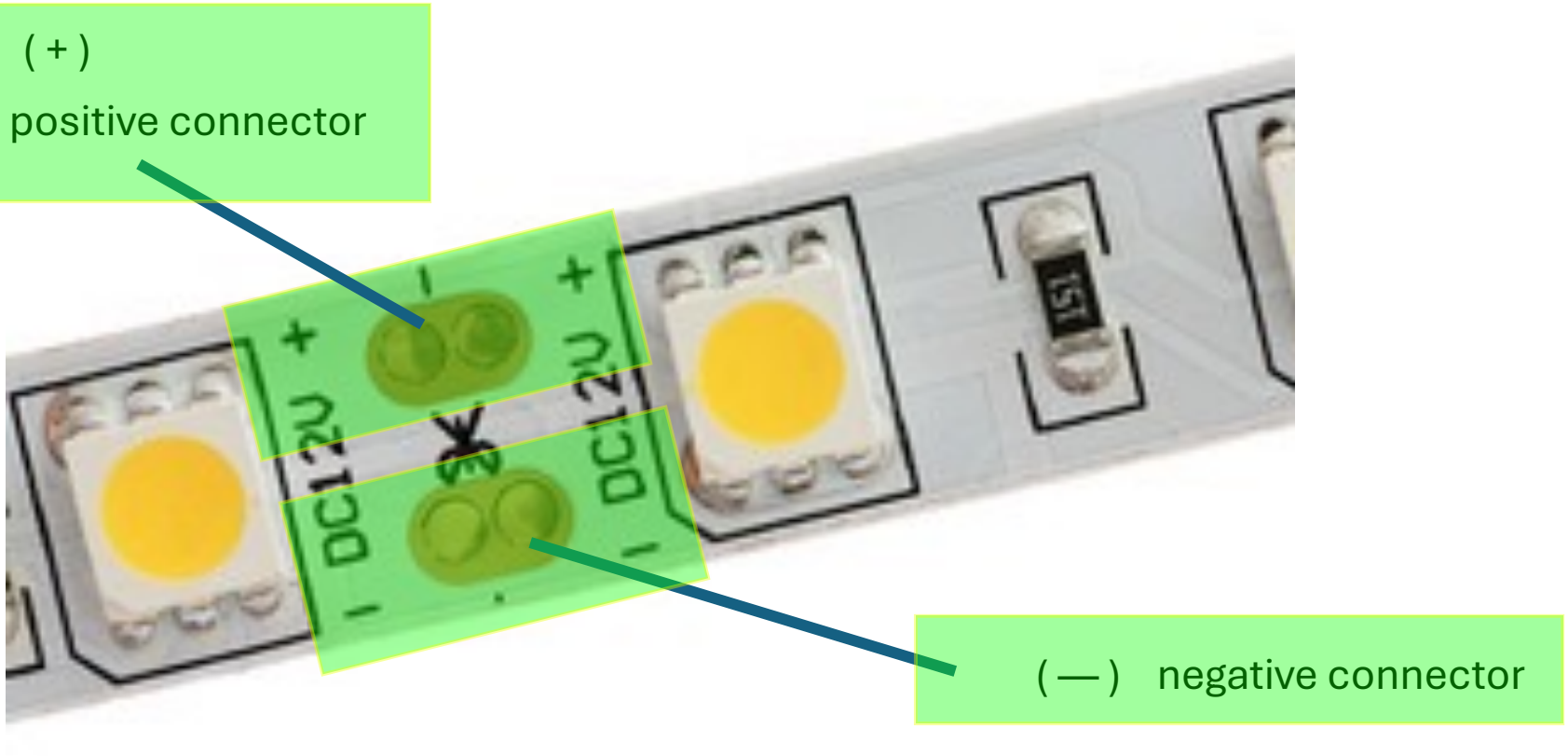
LED strip



Cutline and connection

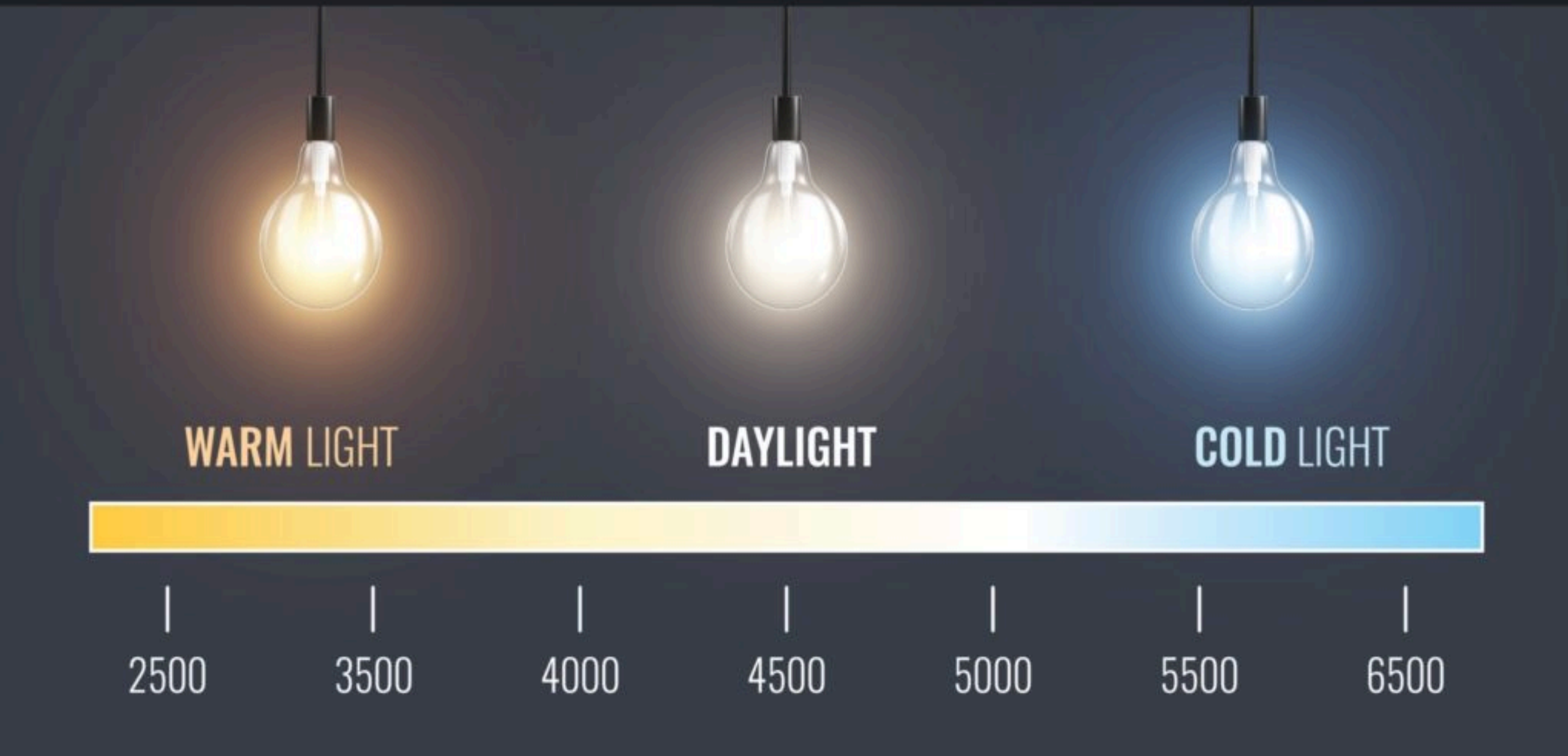
SMD LED strip

LED strip



SMD LED strip

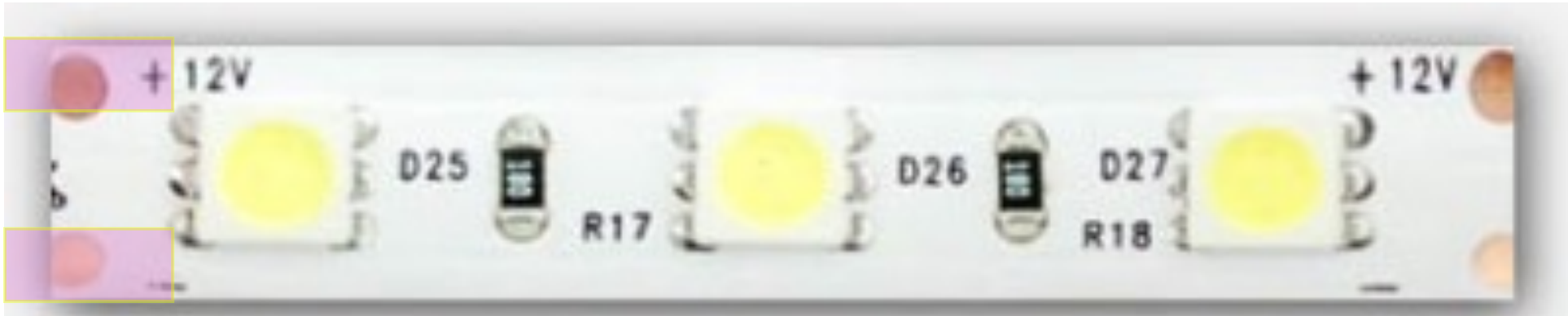
LIGHT TEMPERATURE (K)



Cool / Warm White
LED Strip



White Strip



RGB Strip

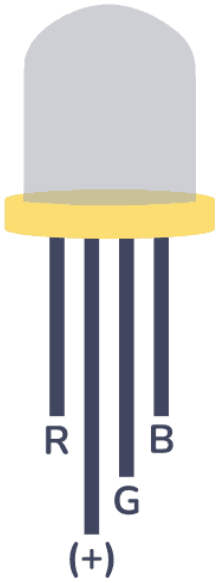


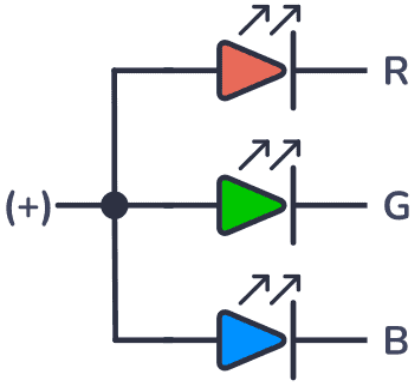
Single Colour

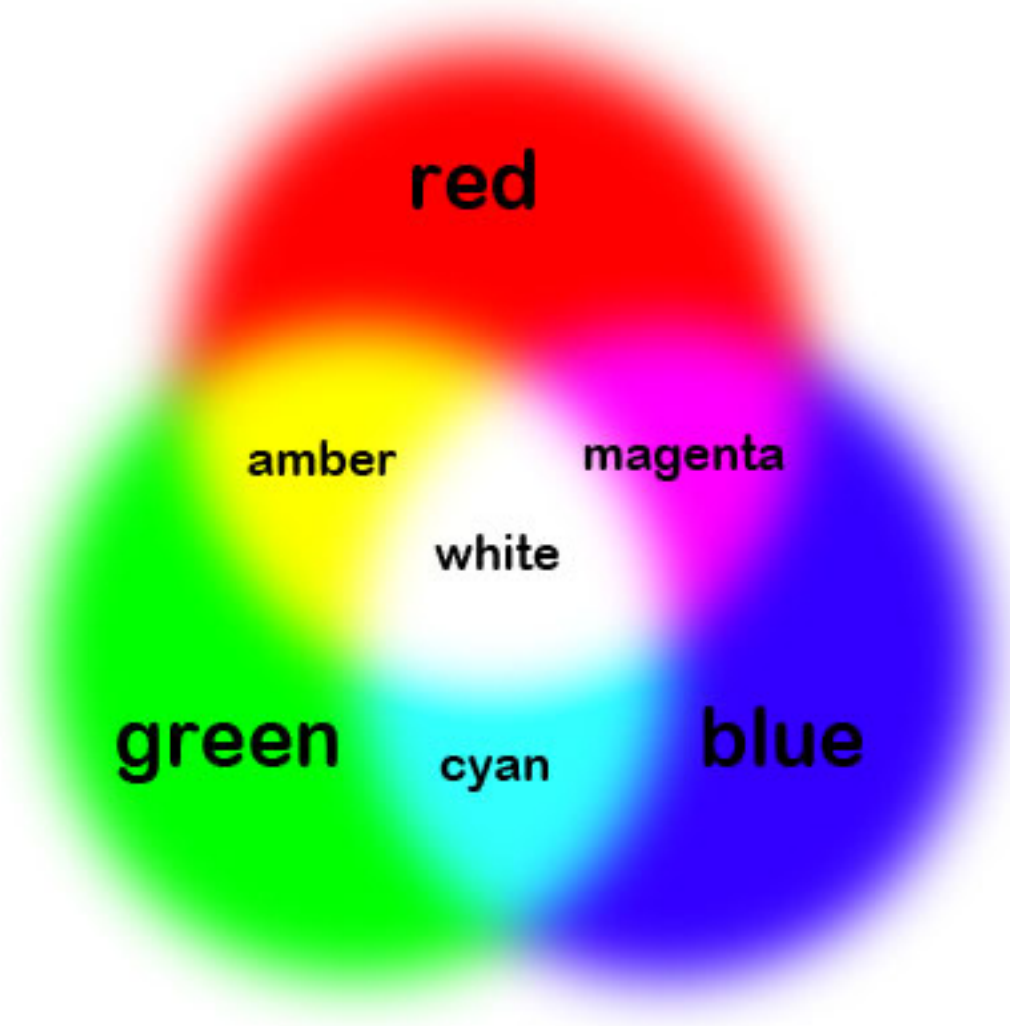


VS

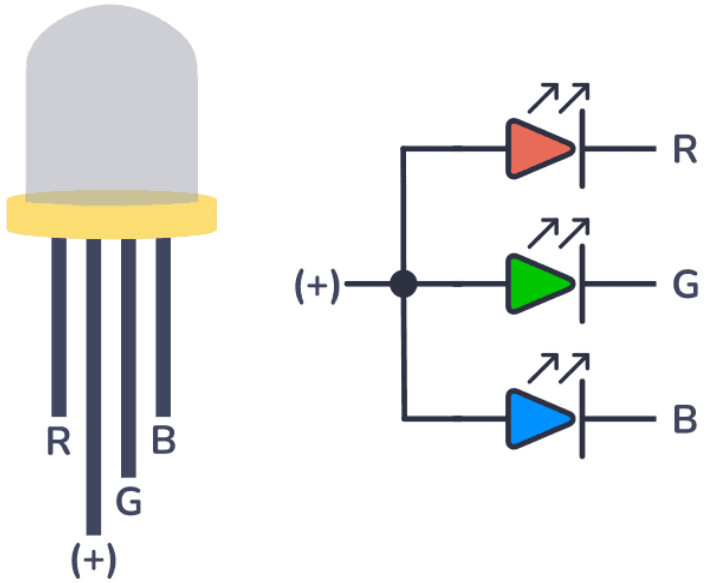
RGB LEDs







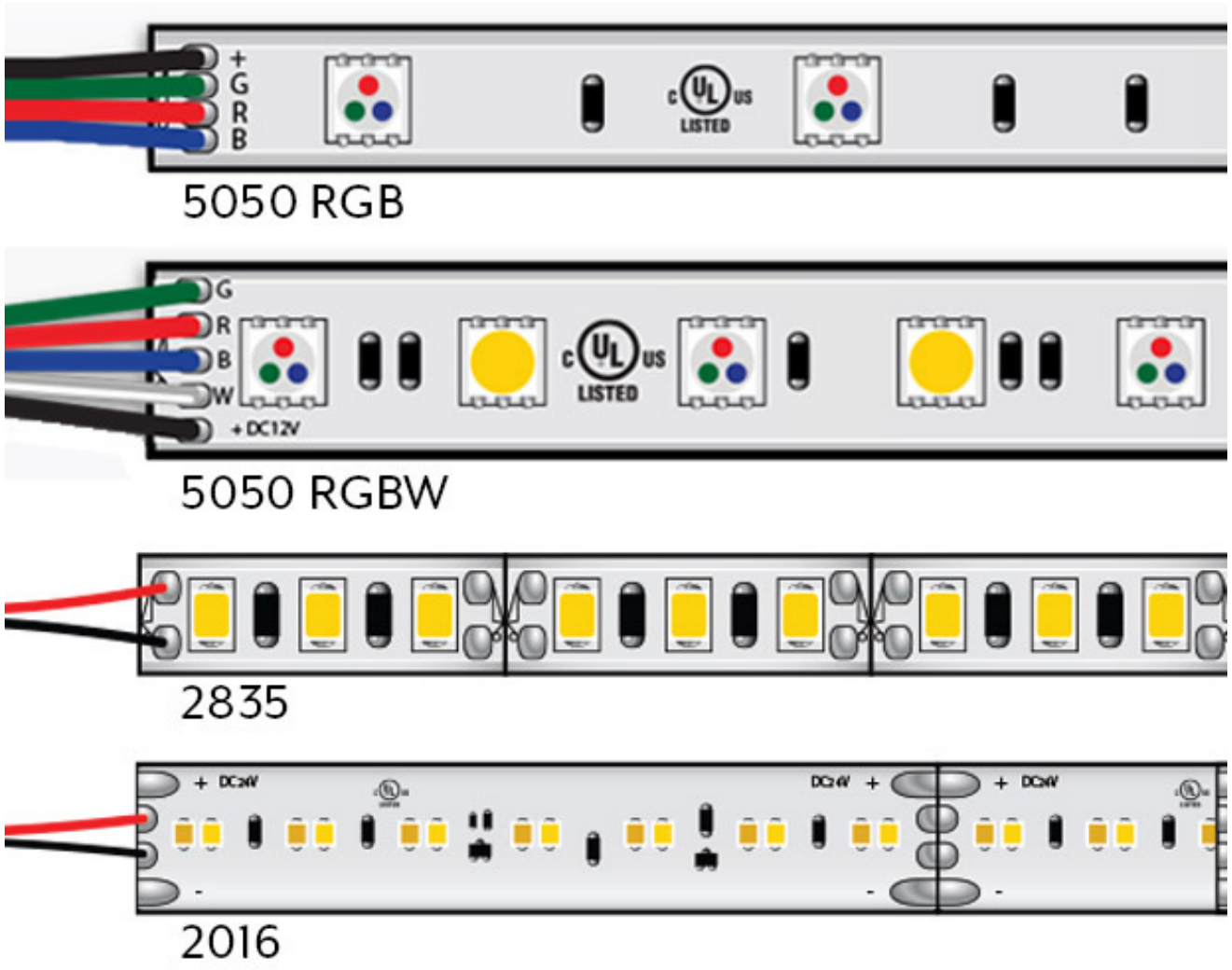
RGB LED STRIP



RGBW LED STRIPs



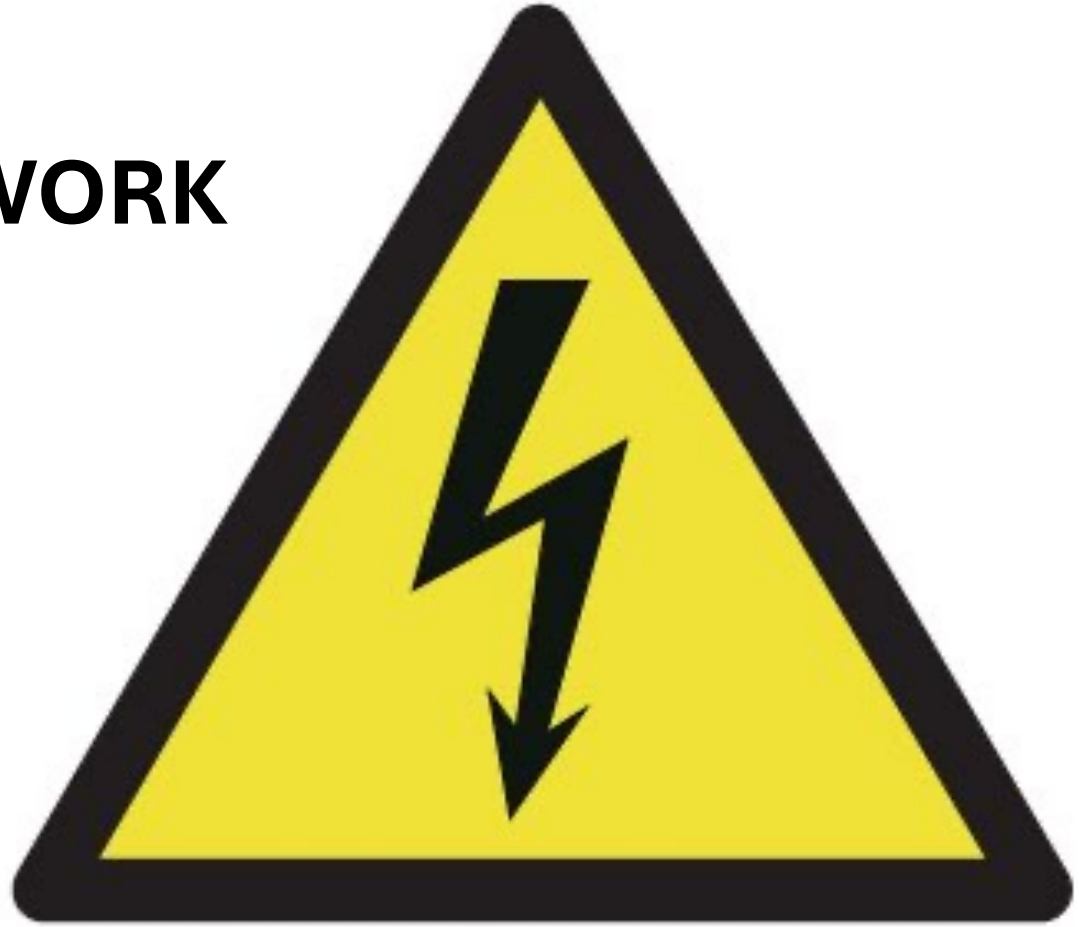
how to choose a LED strip



VOLTAGE

AT hackSpace WECWORK
WITH LOW VOLTAGE
LED STRIPS ONLY

5V-12V



LEDs PER METER



High Density: 36 LEDs per foot (120 LEDs per meter)



Medium Density: 22 LEDs per foot (72 LEDs per meter)

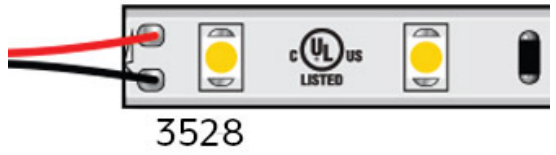


Low Density: 9 LEDs per foot (30 LEDs per meter)

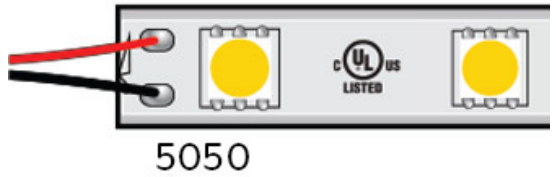
LED SIZE

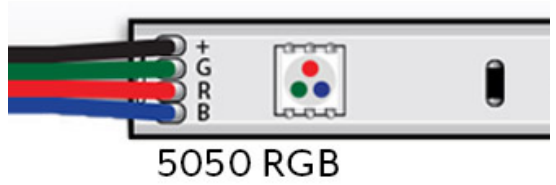
- 5050
- 3528
- 2835

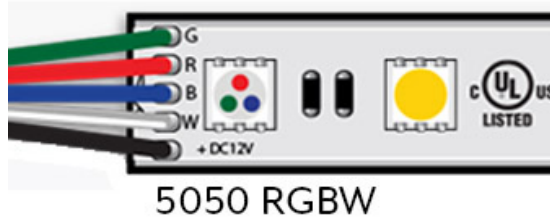
This code only describes the size of the LED component.

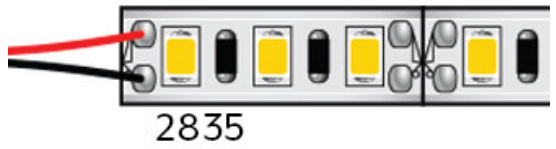


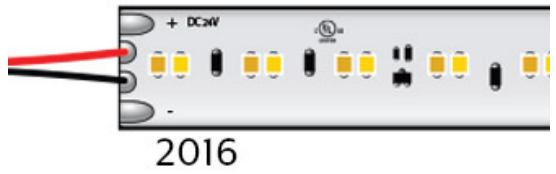














ANALOG VS DIGITAL

- Analog strip, single colour,
  RGB, RGBW, non-addressable

- NeoPixel, Digital LED,
  Pixel LED, Addressable,
  WS2812, WS2811, SK6812



ANALOG VS DIGITAL

- Analog strip, single colour, RGB, RGBW, non-addressable

Analogue = “All LEDs act as one.”

- NeoPixel, Digital LED, Pixel LED, Addressable, WS2812, WS2811, SK6812

Digital = “Each LED can act independently.”

WATERPROOF VS NON-WATERPROOF







IP (Ingress Protection)

- IP20
- **IP65**
- IP67
- IP68

- **Dust (first number, 0-6)**
- **Water (second number, 0-8)**



IP (Ingress Protection)

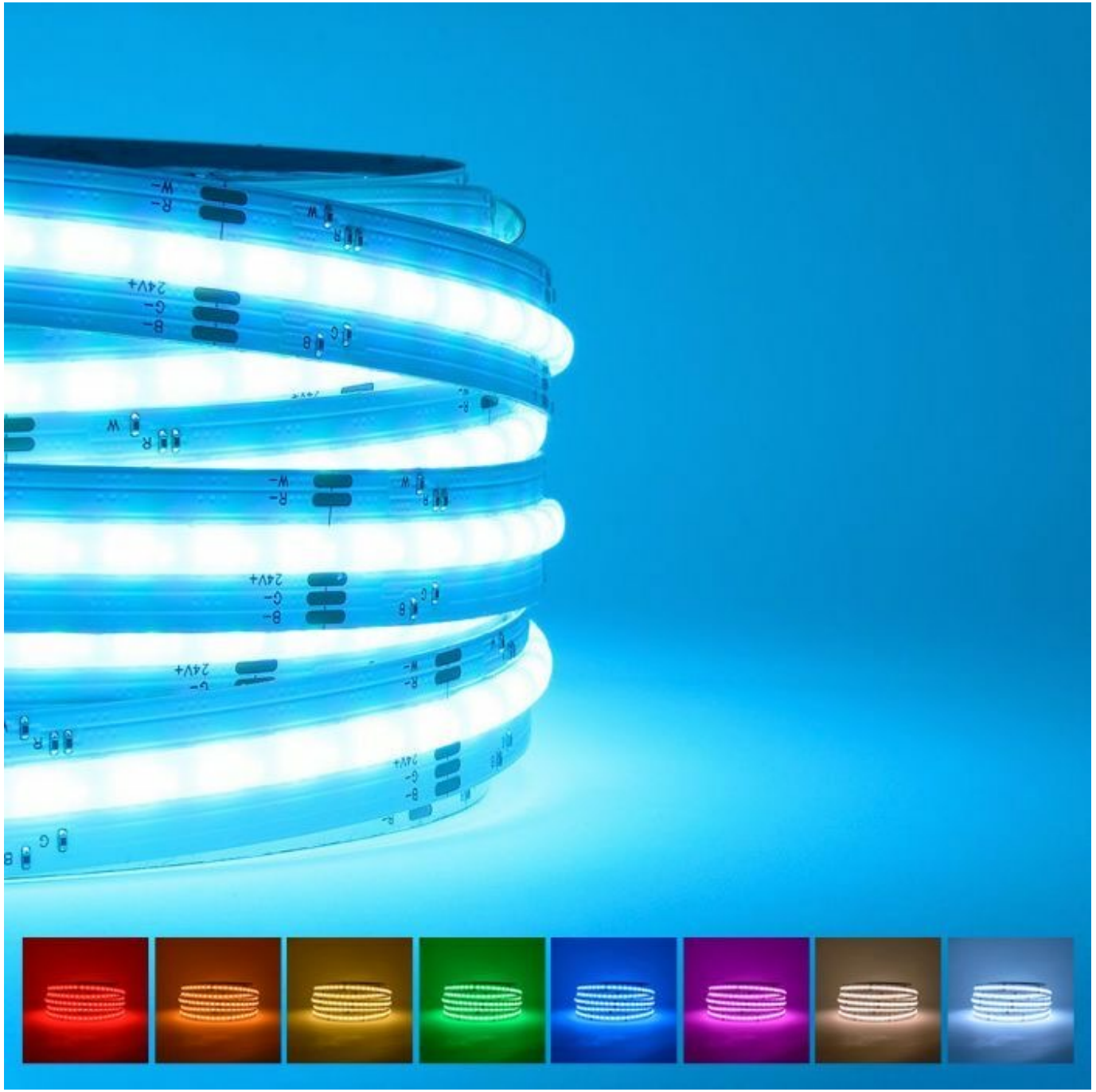
- IP20 (Indoor, no protection)
- **IP65** (Silicone coating, splash resistant but not fully waterproof)
- IP68 (fully sealed, outdoor, waterproof)

COB (Chip On Board) LED STRIPS

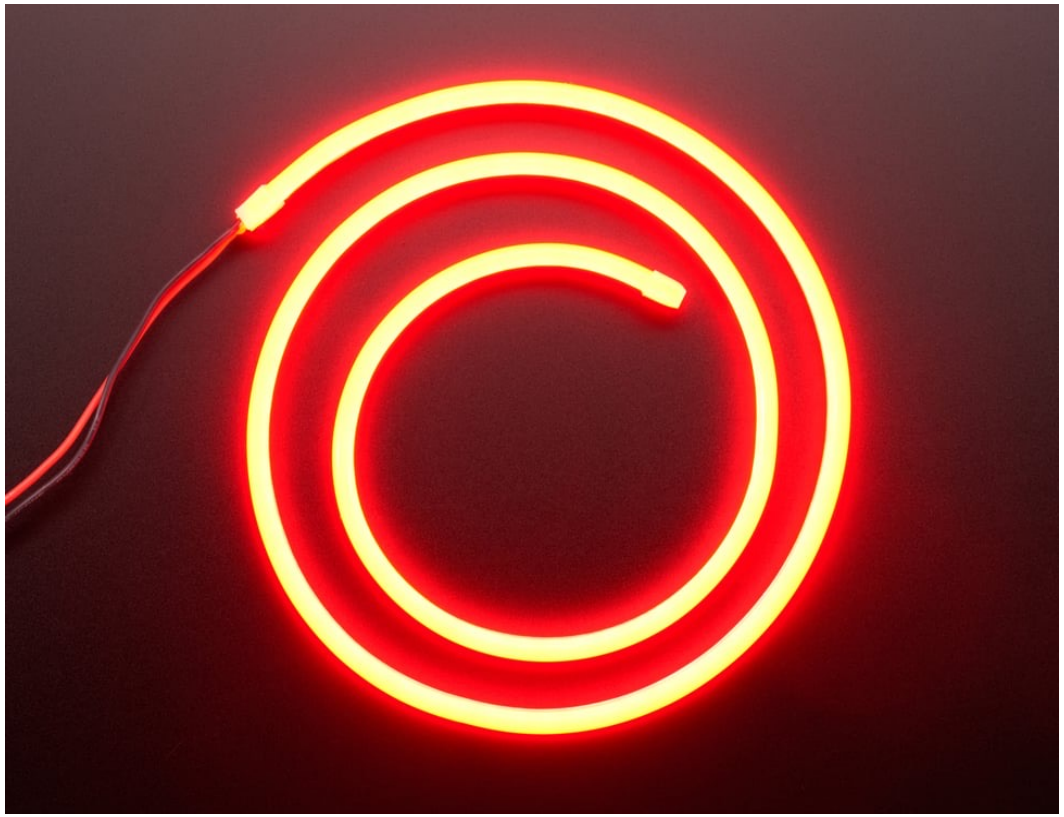
Unlike traditional LED strips, **COB** technology embeds multiple tiny LEDs close together, forming a **continuous line of light** (no visible dots)

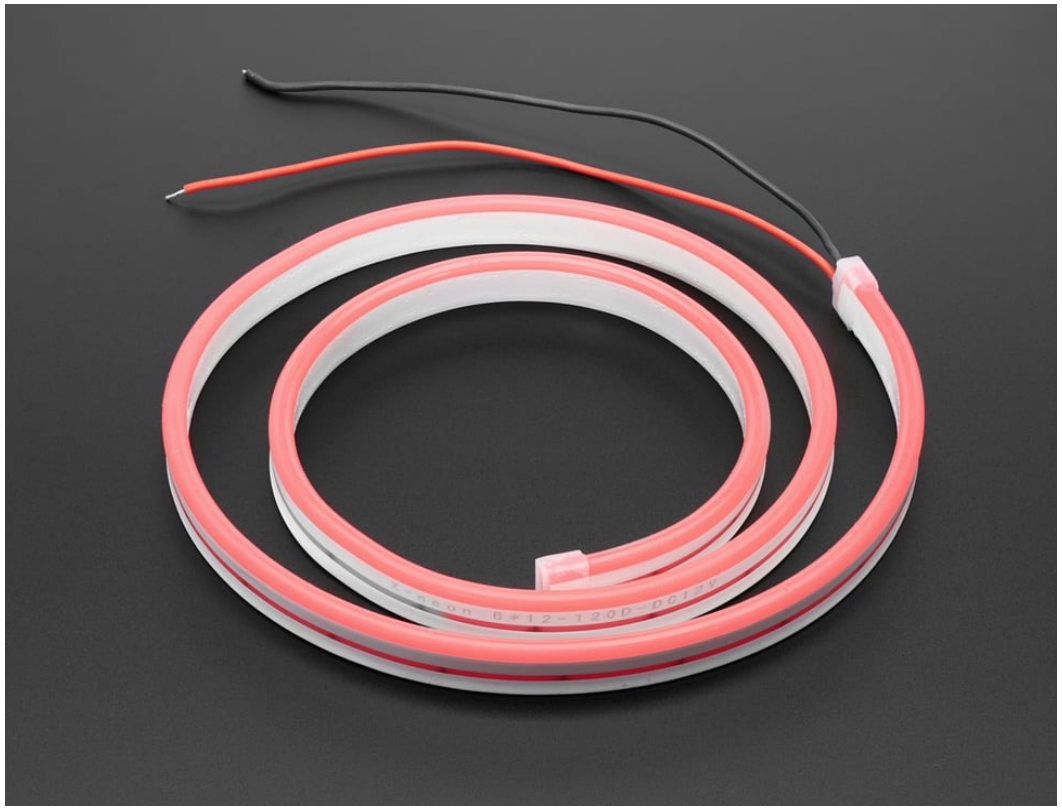
Usually **high-density, bright, and flexible**





COB
Single Colour
or RGB



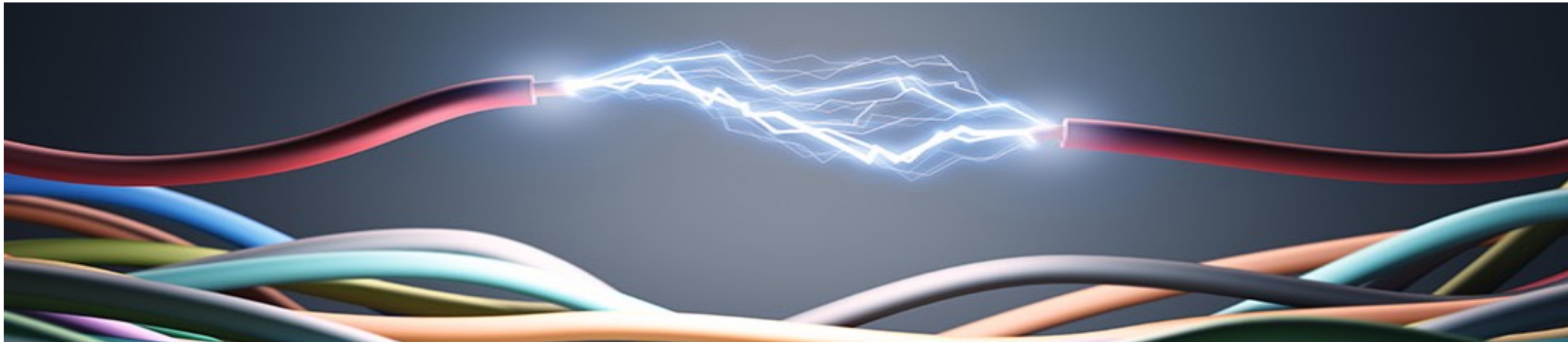


'NEON' LED strips

They are **flexible LED strips embedded in a silicone or PVC tube**. The tube **diffuses the light**, giving a **smooth, continuous “neon” glow**

POWERING

HOW DO I POWER MY LED STRIP?



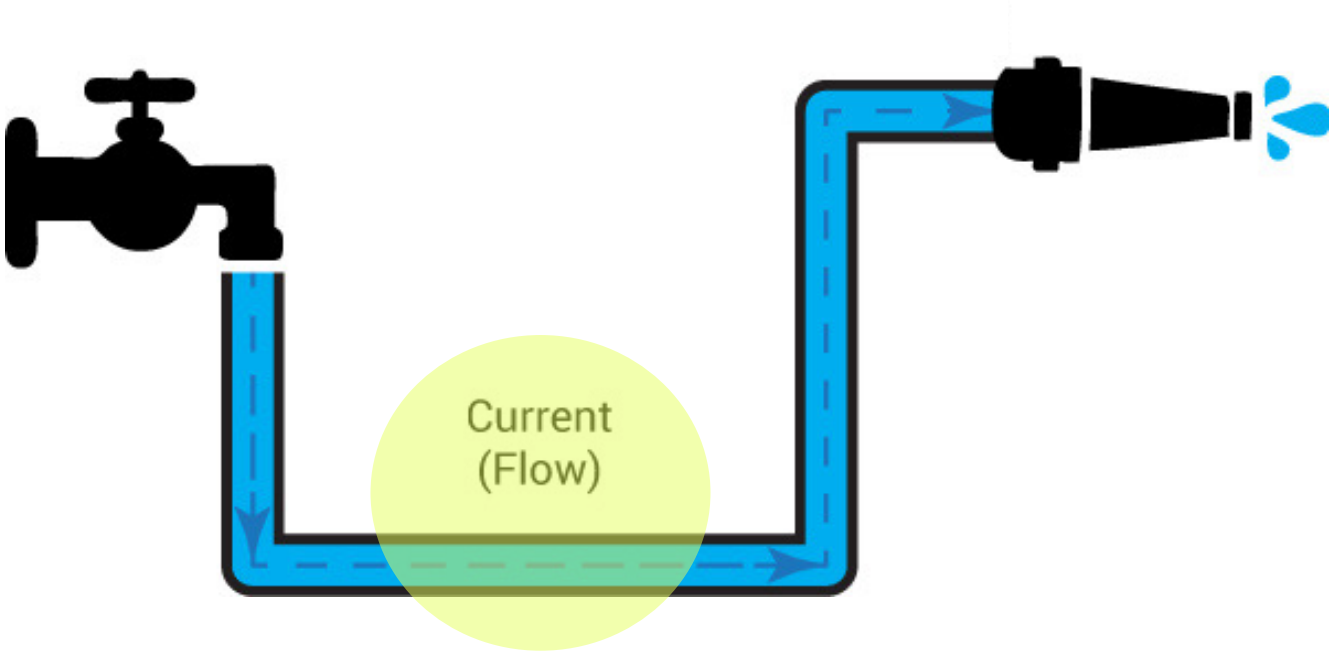
BASIC ELECTRICAL PROPERTIES

POWER	VOLTAGE	CURRENT
watts (W)	voltage (V)	ampers (A)

CURRENT (I)

Current is the amount of electricity flowing in a circuit.

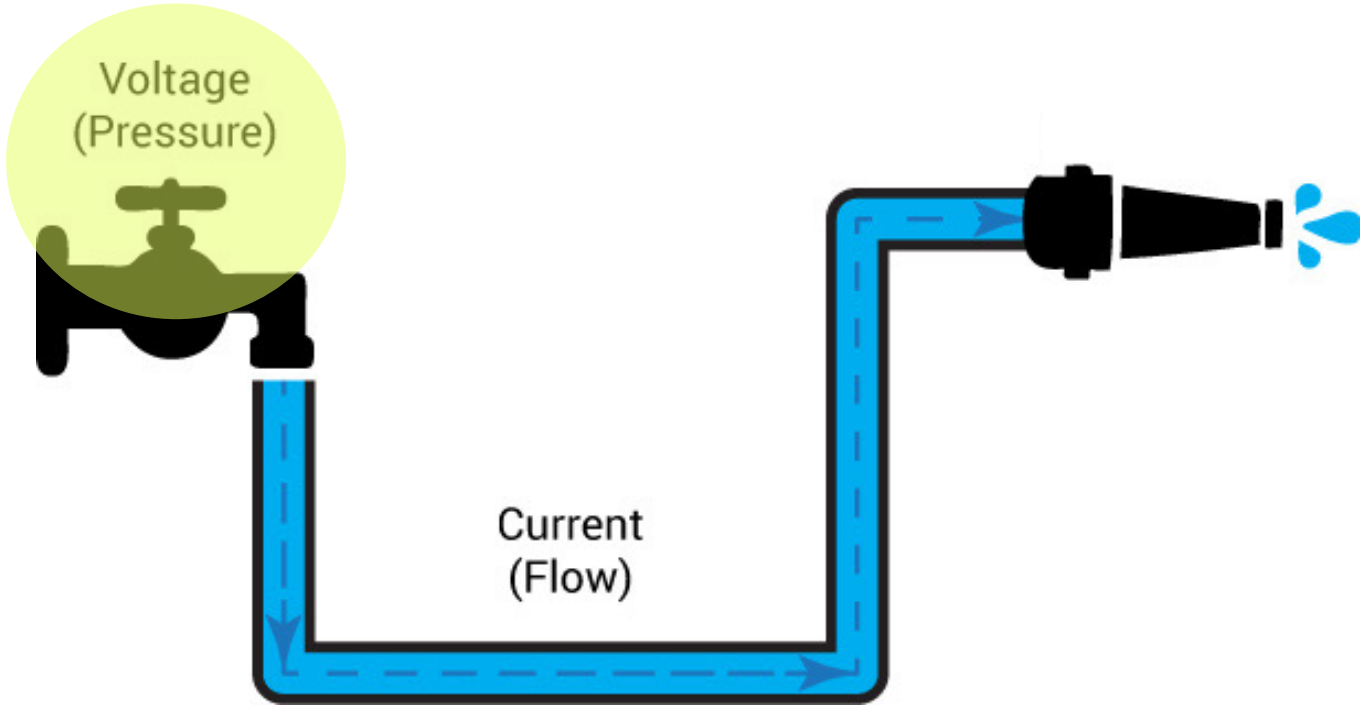
"It is how much electricity being consumed"



VOLTAGE (V)

Voltage is electrical pressure.

"It is the force that pushes electricity through a circuit."

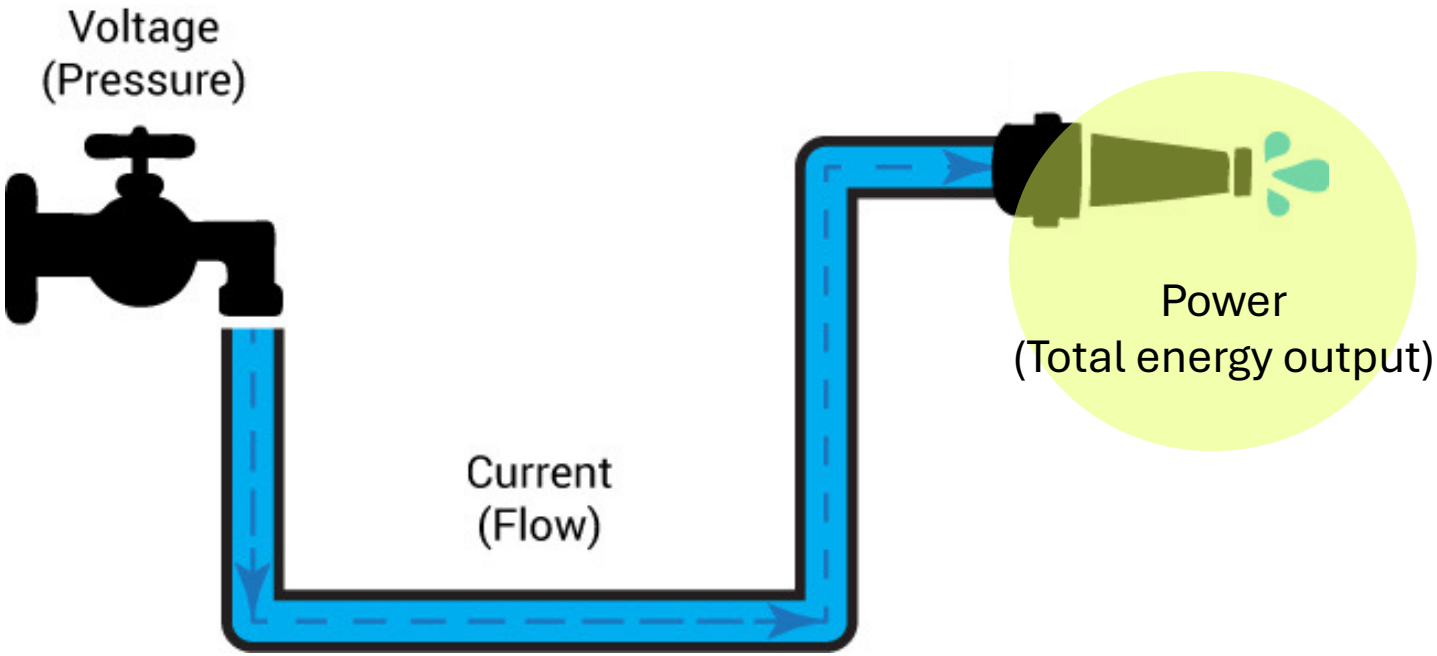


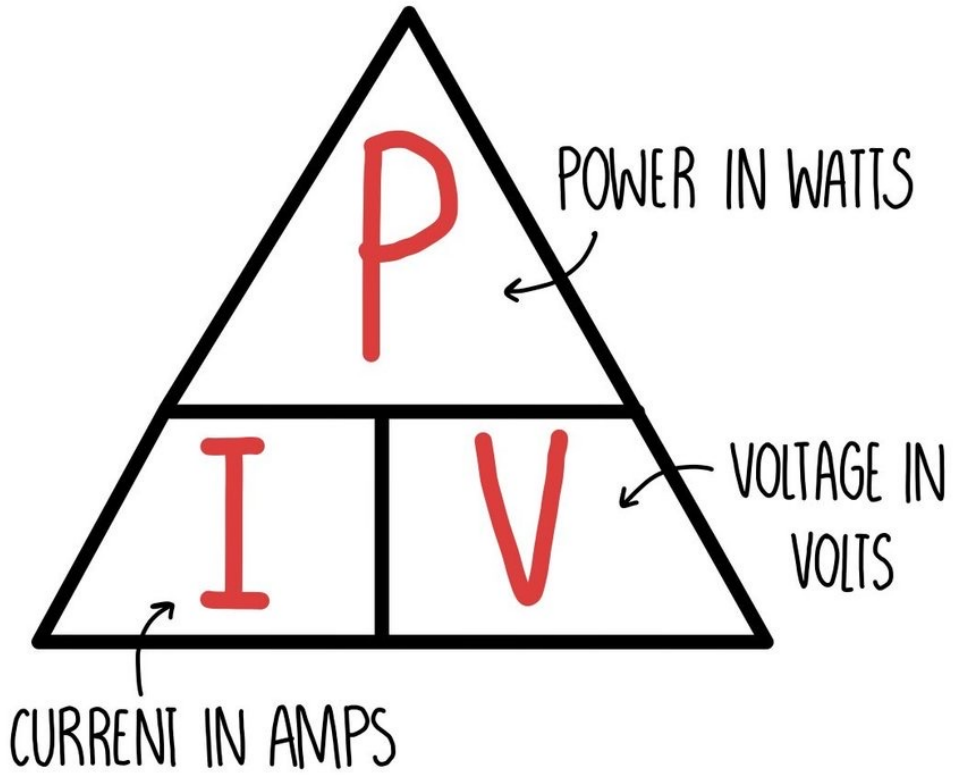
POWER (W)

Power is the product of the Voltage and Current :

$$P = V \times I$$

It tells you the total energy demand of your circuit.





$$P = I \times V$$

$$I = P / V$$

$$V = P / I$$

VOLTAGE

VOLTAGE

12V LED STRIP

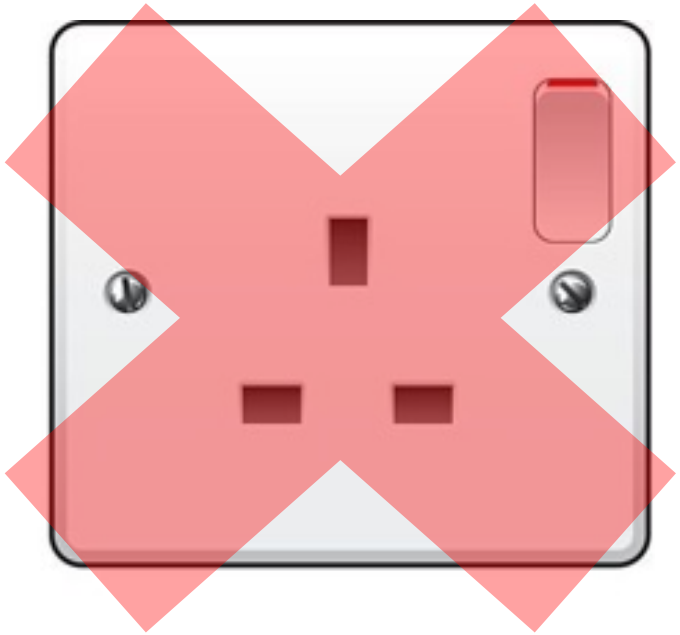


VOLTAGE

12V LED STRIP : 12V POWER SUPPLY

Use the exact voltage — too high will burn the LEDs, too low and they won’t light properly.

Never connect your LED strip directly to the mains



240V

VOLTAGE

You will need a
12V TRANSFORMER

POWER







CIE xy

Downl

CTRICAL SPECIFICATIONS

Input type:	Constant Voltage
Input voltage:	12V DC
Current draw per ft:	490 mA @ 12V DC
Current draw per reel:	7.5 A @ 12V DC
Power draw per ft:	5.5 W @ 12V DC
Power draw per reel:	90 W @ 12V DC
Max run:	16.4 ft (5 meters)

VER SUPPLY SELECTION

with Waveform Lighting PN 3091 or

a third-party p

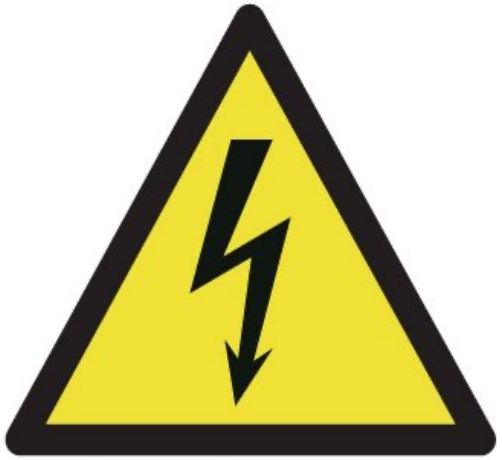
OUTPUT

=

12V



- HIGH VOLTAGE IS DANGEROUS!
- Only work with the power supply provided in this course.
- Never cut or modify mains wiring



CURRENT

Voltage stays constant along the strip — every LED gets the same 12V.

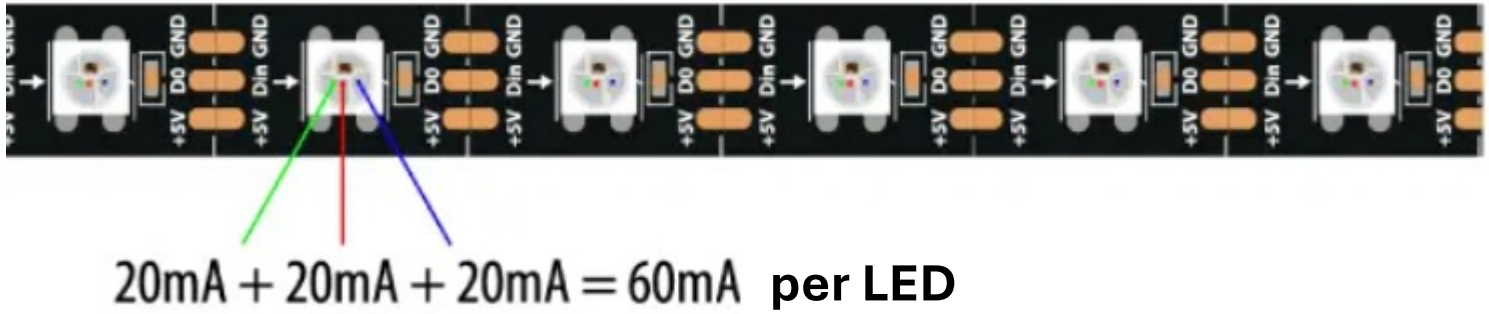
Current is additive — every LED you add draws its own current, and those all add up at the power supply end.

More LEDs = more current demanded.



WHITE LED STRIP

20 mA per LED



RGB LED STRIP

per LED

Warning! This values are general standards but may vary depending on the LED strip. Always check the powering values of your specific LED strip for safety.



20 mA per LED

(20 mA = 0.02 A)

Example: 20 white LEDs

0.02A	X	Number of LEDs

= total current



20 mA per LED

0.02A x 50 LEDs = 1 A



20 mA per LED

0.02 A x 50 LEDs = 1 A

1A / 0.8 = 1.25 A

Power Supply must be 12V, 1.25A (or higher)

+20% rule

The Power Supply CURRENT rating should always be at least 20% higher than the total current draw of your LEDs.

...Power Supply must be 12V, 1.25A (or higher)

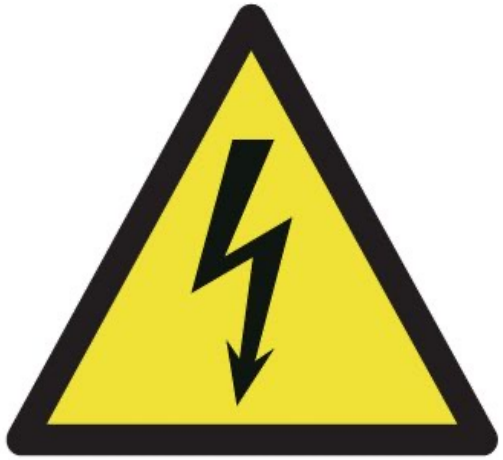
12V 0.5A

12V 1A

12V 2A

ETC...

CURRENT



- CAN BE DANGEROUS IF NOT USED PROPERLY.
- TOO MUCH OR INCORRECT CURRENT CAN CAUSE ELECTROCUSION, OVERHEAT, FIRE.

POWER

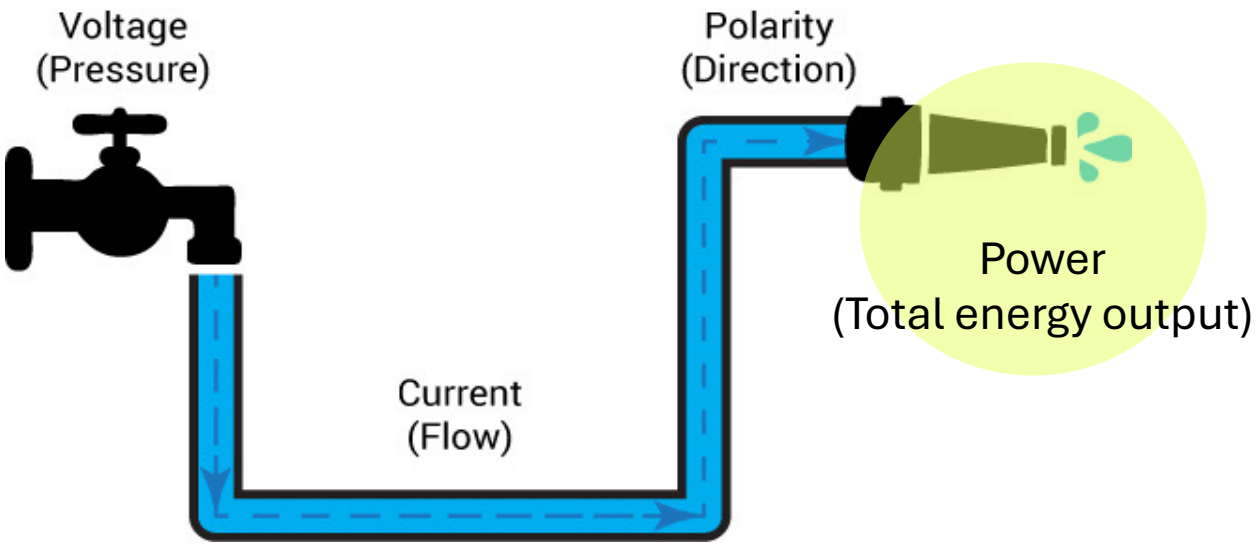


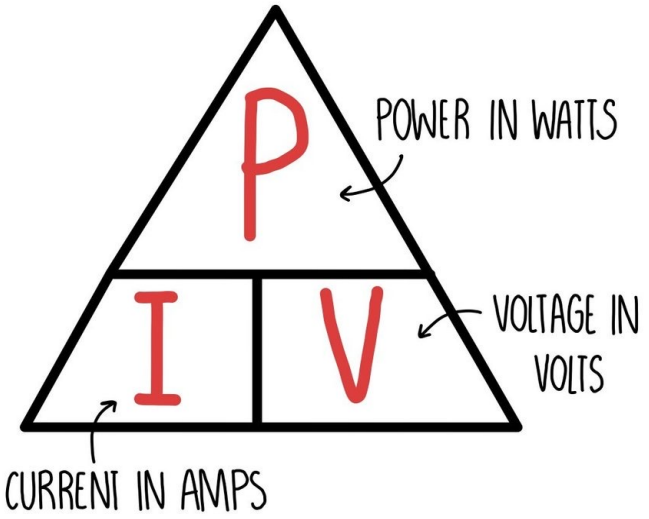
Manufacturer	ATOM LED
Product Type	LED Strip Light
Colour	RGB
IP Rating	IP67 Waterproof
LED per mtr	60 p/Mtr
Metres	5 Metres
Guarantee Period	2 Years
Wattage	70w
Size	14mm x 5mm
Input Voltage:	DC 12V
Brightness	High Brightness
Reel Length	5m
Beam Angle:	120 degree

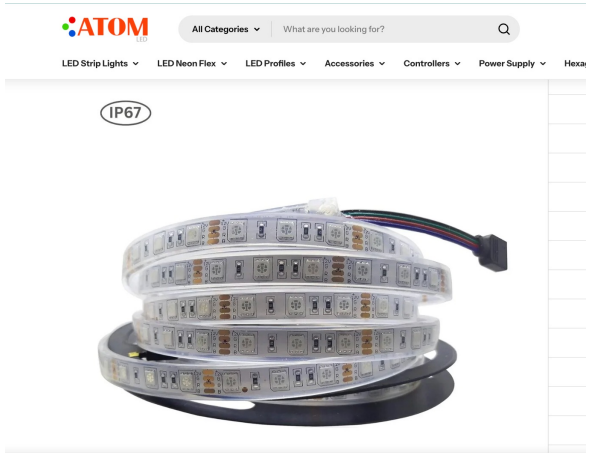
POWER (W)

Power is the product of the Voltage and Current :

$$P = V \times I$$







12V LED STRIP

5 meters

70 Watts

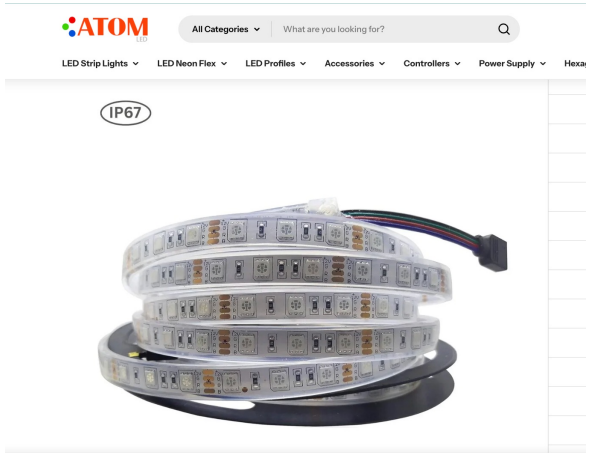
60 LEDs per METER

300 total LEDs

POWER SUPPLY

Should be....

Voltage: ?



12V LED STRIP

5 meters

70 Watts

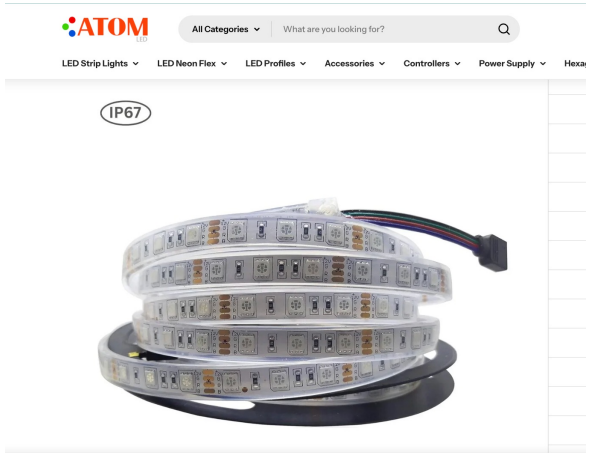
60 LEDs per METER

300 total LEDs

POWER SUPPLY

Should be....

Voltage: 12V



12V LED STRIP

5 meters

70 Watts

60 LEDs per METER

300 total LEDs

POWER SUPPLY

Should be....

Voltage: 12V

Power: ?

+20% rule

The Power Supply WATTAGE rating should always be at least 20% higher than the total power draw of your LEDs.

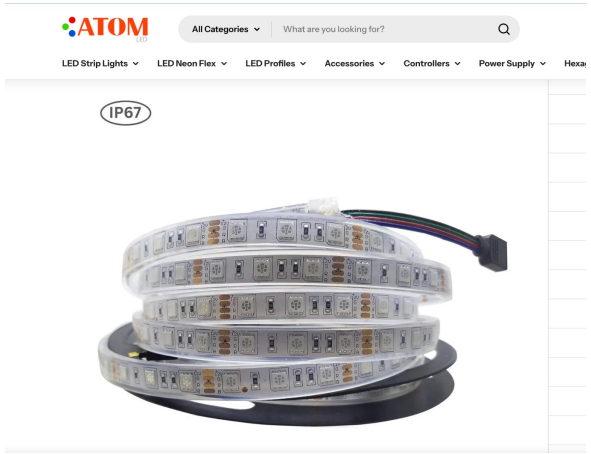
Example:

If my LED strip has
a total power draw (5 meters) of 70W...

70W ÷ 0.8 = **87.5W ....**

add 20% margin

Power Supply should provide 90W or higher



12V LED STRIP

5 meters

70 Watts

60 LEDs per METER

300 total LEDs

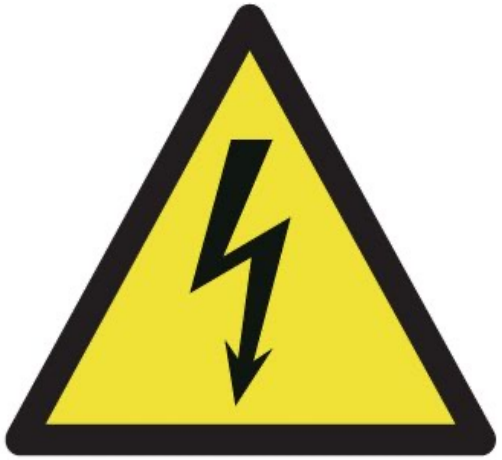
POWER SUPPLY

Should be....

Voltage: 12V

Power: 90W or higher

Powering for beginners and hobbyists
(not professional electricians)



- **A safe threshold of 12V 50W** power supply (equivalent to 12V 5A) is advised.

- Massive power supplies to power massive electrical project could lead to serious electrical hazards.

- Consult an electrician for high voltage/power projects.



WARNING:

This course doesn't cover high voltage neither powering large meters of LED strips.

Long LED strips require specific power distribution setups to avoid **voltage drop** and **overheating**.

⚠️ These advanced power setups will **not be covered in this course**.
In this class, we will work with **short, low-voltage sections only** for safety.

POWER SUPPLIES

TRANSFORMER



LED DRIVER



Transformer or LED-driver?

- Any that can reliably provide constant voltage.

- For prototyping and learning, a transformer is enough.

- For **finished installations, public-facing work, or anything permanent** — a proper LED driver is the right tool. **Continuous operation** — LED installations run for hours, the driver is rated for that.

Transformer or LED-driver?

- Any that can reliably provide constant voltage.



prototyping
and
learning



For **finished installations, public-facing work, or anything permanent** — a proper LED driver is the right tool.

Continuous operation — LED installations run for hours, the driver is rated for that.

LED DRIVER





ACTIVITY

How would you power your LED strip?

12V LED Strip,
1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

Voltage :

Power:

Amperage:

ACTIVITY

How would you power your LED strip?

12V LED Strip,
1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

Voltage : 12V

Power:

Amperage:



ACTIVITY

How would you power your LED strip?

12V LED Strip,
1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

Voltage : 12V

Power: ?

Amperage:

POWER: ?

LED STRIP: 1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

POWER: ?

LED STRIP: 1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

TOTAL POWER = 30W (for 60 LEDs)

POWER: ?

LED STRIP: 1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

TOTAL POWER = 30W (for 60 LEDs)

30W / 60 LEDs =

POWER: ?

LED STRIP: 1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

TOTAL POWER = 30W (for 60 LEDs)

30W / 60 LEDs = 0.5W per LED

POWER: ?

LED STRIP: 1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

TOTAL POWER = 30W (for 60 LEDs)

30W / 60 LEDs = 0.5W per LED

0.5W x 10 LEDs =

POWER: ?

LED STRIP: 1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

TOTAL POWER = 30W (for 60 leds)

30W / 60 LEDs = 0.5W per LED

0.5W x 10 LEDs = 5W ... plus 20% margin

POWER: ?

LED STRIP: 1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

TOTAL POWER = 30W (for 60 leds)

30W / 60 LEDs = 0.5W per LED

0.5W x 10 LEDs = 5W ... plus 20% margin

5W / 0.8 = 6.25W or higher



ACTIVITY

How would you power your LED strip?

12V LED Strip,
1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

Voltage : 12V

Power: 6.25W or higher

Amperage:

ACTIVITY

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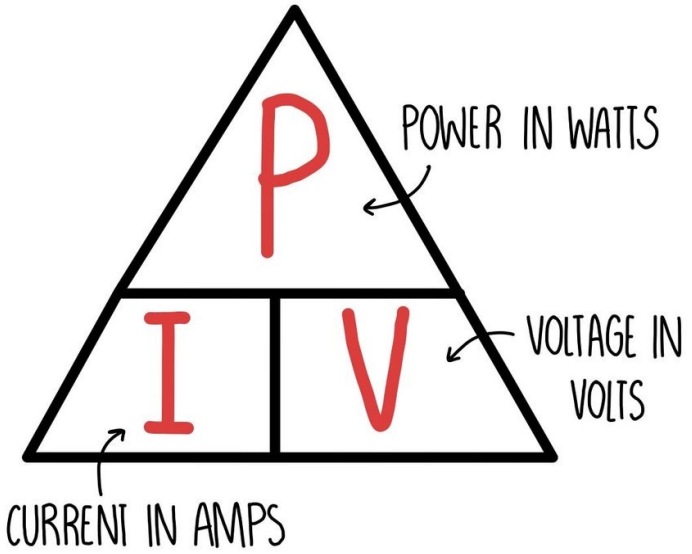
Amperage: ?

Current: ?

Voltage : 12V

Power: 6.25W or higher

Amperage: ?



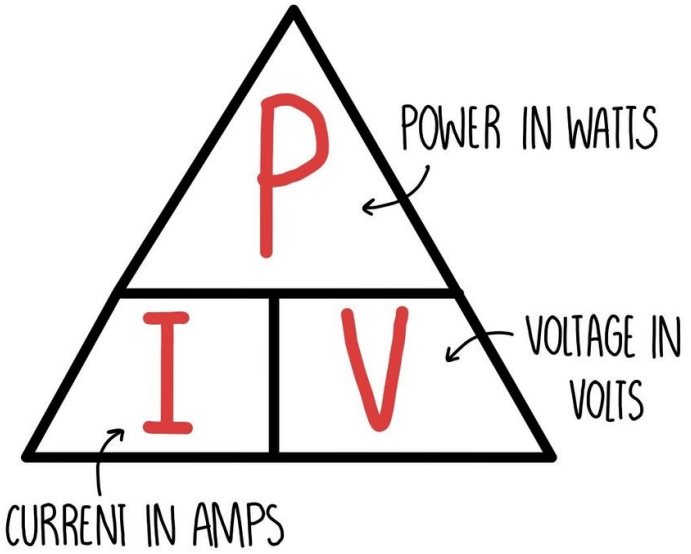
Current: ?

Voltage : 12V

Power: 6.25W or higher

Amperage: ?

I = P / V



Current: ?

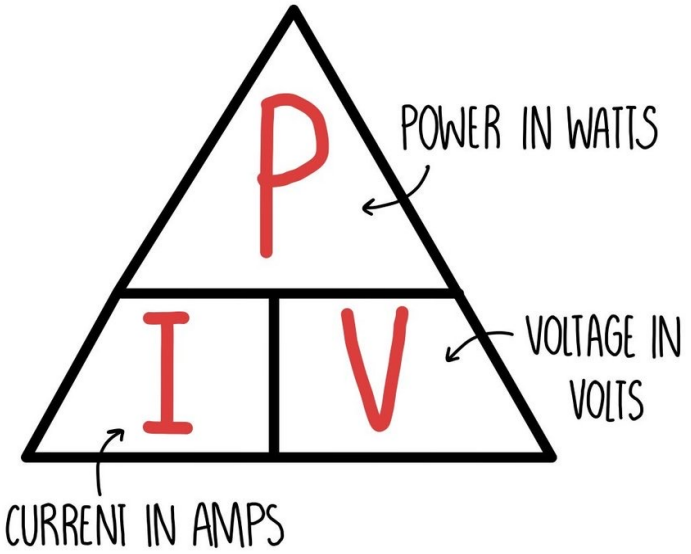
Voltage : 12V

Power: 6.25W or higher

Amperage: ?

I = P / V

I = 7W / 12V



Current: ?

Voltage : 12V

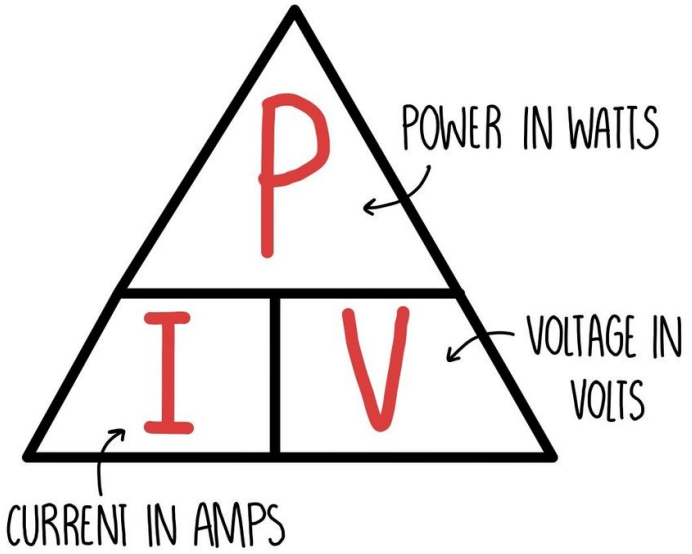
Power: 6.25W or higher

Amperage: ?

I = P / V

I = 7W / 12V

I = ~0.58A or higher



ACTIVITY

How would you power your LED strip?

12V LED Strip,
1 meter, 60 LEDs per meter, 30W

1 - If you only want to use 10 LEDs ....

Voltage : 12V

Power: 6.25W or higher

Amperage: 0.58W or higher

Voltage : 12V

Power: 6.25W or higher

Amperage: 0.58A or higher

12V 1A Power Supply



ADVICE:

DO NOT buy cheap equipment.

Buy equipment with clear powering details and good-quality manufacture.



FURTHER LEARNING

A Complete Guide to: The Light Emitting Diode
https://www.electronics-tutorials.ws/diode/diode_8.html

LEDs
https://learn.sparkfun.com/tutorials/light-emitting-diodes-leds

More about LED strips
https://www.tweaking4all.com/hardware/arduino/arduino-ws2812-led

Electricity Basics
https://learn.sparkfun.com/tutorials/voltage-current-resistance-and-ohms-law

Electrical Power
https://learn.sparkfun.com/tutorials/electric-power

How to use a multimeter
https://learn.sparkfun.com/tutorials/how-to-use-a-multimeter

LED Drivers
https://uk.rs-online.com/web/content/discovery/ideas-and-advice/led-drivers-guide?msockid=261b4ec2fa2a669704175a17fb9167d1